

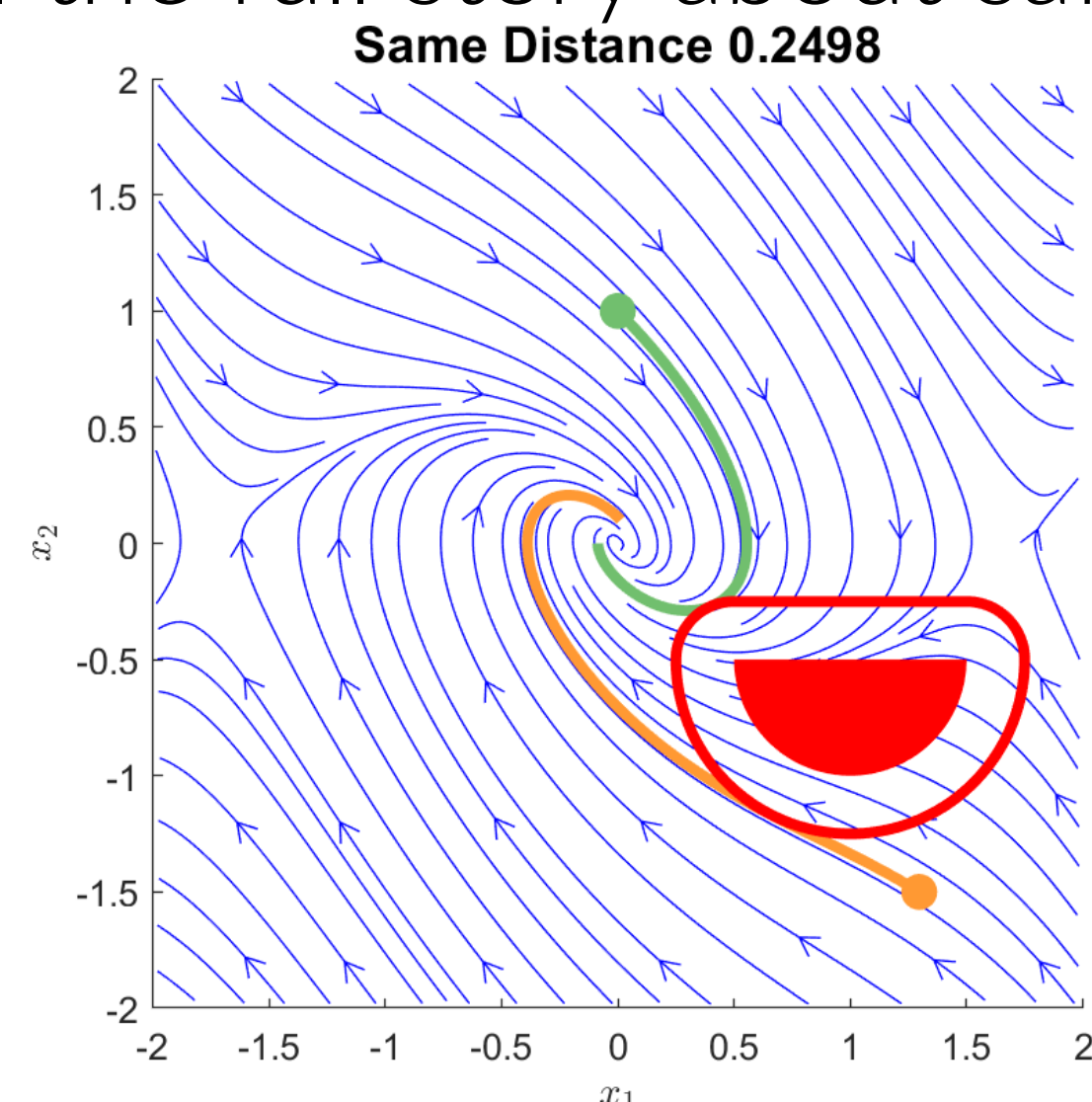
How Much Data Corruption is Required to Crash?

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Why Corruption?

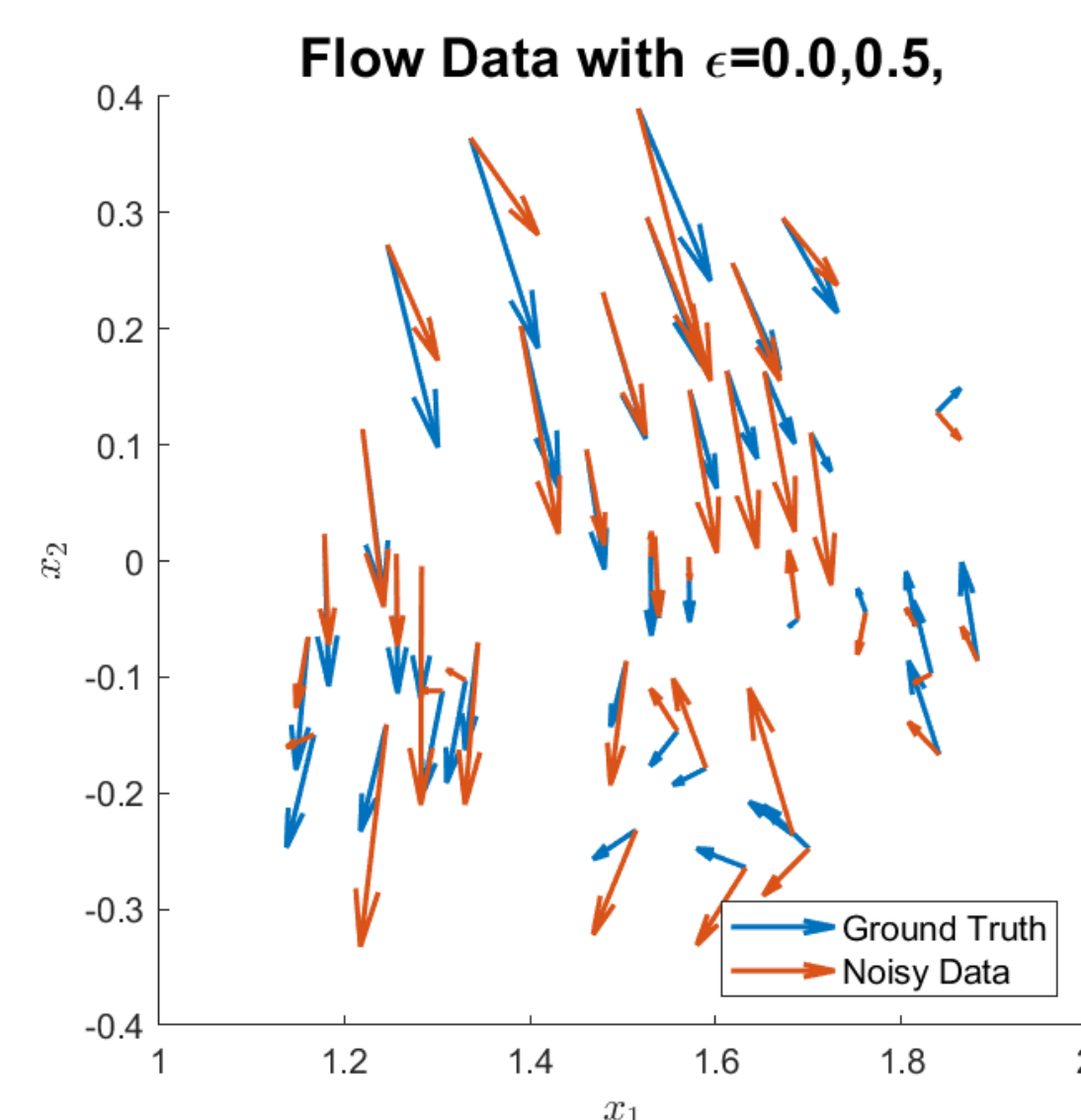
The distance to an unsafe set X_u does not tell the full story about safety



Green takes 2× control effort as yellow to crash (more corruption).

Corrupted Data

Noisy data $\mathcal{D} = \{(t_k, x_k, y_k)\}_{k=1}^{N_s}$ gathered from unknown $\dot{x} = F(t, x)$
 $\forall k = 1..N_s \quad \|y_k - F(t_k, x_k)\|_\infty \leq \epsilon.$



Corruption Function

A dictionary $(f_0, \{f_\ell\})$ is given with

$$f(t, x; w) = f_0(t, x) + \sum_{\ell=1}^L w_\ell f_\ell(t, x)$$

w are parameters of system model

Per-sample corruption function $J_k(w)$

$$\|f_0(t_k, x_k) + \sum_{\ell=1}^L w_\ell f_\ell(t_k, x_k) - y_k\|_\infty$$

Joint-data corruption of $\mathcal{D}$:

$$J(w) = \max_{k=1..N_s} J_k(w)$$

Crash OCP

Minimize the corruption necessary to crash into unsafe set X_u .

Start at $x_0 \in X_0$, follow $\dot{x} = f(t, x(t); w)$ until $x(t^* | x_0, w) \in X_u$.

Z^* : infimum of z w.r.t. trajectory with
 $z \geq J(w), \dot{z} = 0$

Nonconvex Mayer-form OCP in the variables (t^*, x_0, z, w)

Linear Program

Infinite-dimensional convex program (HJB subsolution):

$$\begin{aligned} q^* &= \sup_{\gamma \in \mathbb{R}, v} \gamma \\ \text{in } X_0 : & \quad \gamma \geq v(0, x, z) \\ \text{in } X_u : & \quad v(t, x, z) \leq z \\ \text{if } z &\geq J(w) : \mathcal{L}_f v(t, x, z, w) \geq 0 \\ & v(t, x, z) \in C^1 \end{aligned}$$

Valid bound $Z^* \geq q^*$, conservative w.r.t. time-dependence of $w(t)$

Complexity

Use moment-SOS/neural net/gridded LPs to get lower bounds on q^*

All suffer curse of dimensionality

Robust counterparts can eliminate w (treated as external input)

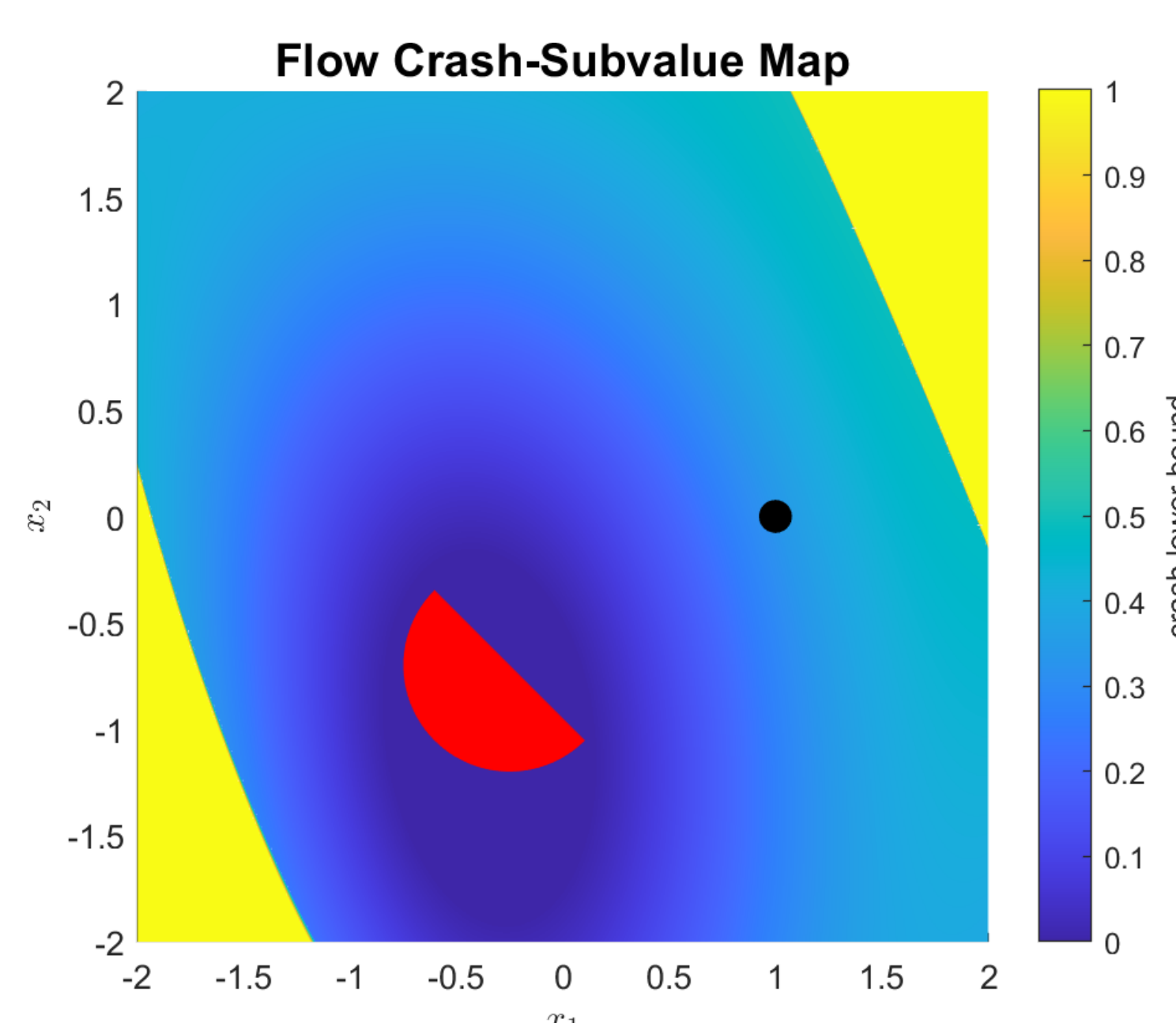
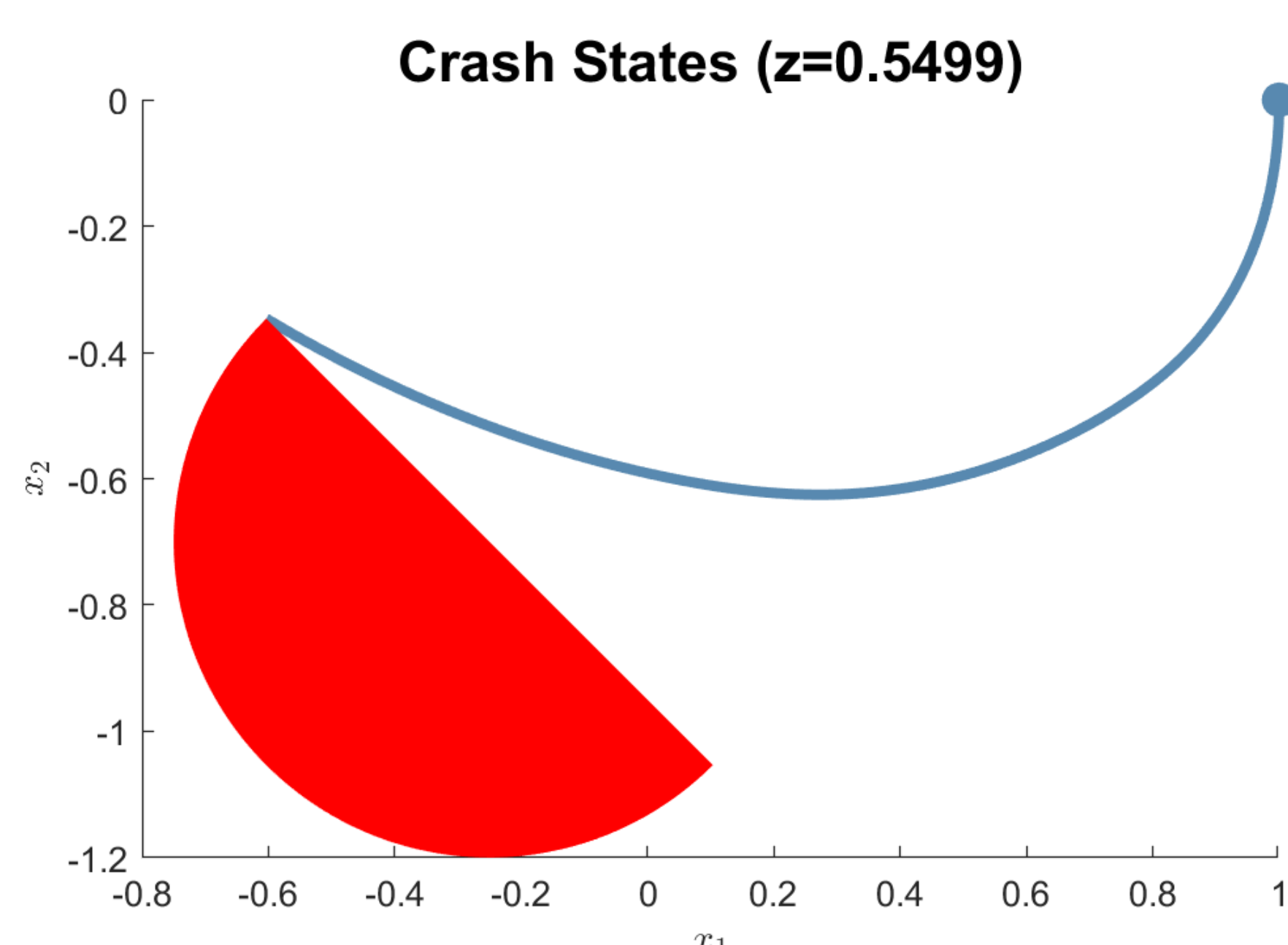
Scaling of $\mathcal{L}_f v \geq 0$ constraint:

Method	Variables	Count
Original	(t, x, z, w)	$n + 2 + L$
Robust	(t, x, z)	$n + 2$

Crash-Safety Example

Known that $\dot{x}_1 = x_2$, $\dot{x}_2$ is a cubic polynomial ($L = 10$) Compare crash and distance bounds to X_u (moment-SOS truncation):

$F(t, x) = \begin{bmatrix} x_2 \\ -x_1 - x_2 + \frac{1}{3}x_1^3 \end{bmatrix}$	Order	1	2	3	4
	Crash	0.0582	0.4423	0.4864	0.5499
	Distance	1.698×10^{-5}	0.1936	0.2003	0.2009



Slightly more data corruption will cause the system to crash
 $(\epsilon = 0.5, Z^* = 0.5499)$

Conclusion

System unsafety can be quantified by control effort needed to crash

Specifically, bound the minimum **data corruption** needed to crash

Extensions:

- Discrete-time dynamics
- Hybrid systems
- Online verification
- Semidefinite-bounded uncertainty

